

Project Title
Automated Log Error Monitoring System

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms - 3 (25CS2103E)

in

A.Y. 2026-2027

Submitted by
G. Eesha Yuktha (2520030436)
Varshini Bolla (2520030431)
Sai Saranya (2520030258)

Team Number: 18
Section: 5

Under the guidance of

D. Swathi
CSE



Hyderabad-500090, Telangana, India.

ABSTRACT

Modern servers generate large volumes of log data, making it difficult for system administrators to manually identify and analyse errors. This project proposes a log analysis system that detects recurring error patterns from massive server log files. The system uses efficient string matching algorithms to search for predefined error signatures and identify common errors. Detected errors are categorised based on their frequency and severity, making it easier to understand the most important issues.

The system also supports continuous monitoring of log files, allowing newly generated errors to be detected in real time. It generates summary reports containing useful information such as error type, occurrence frequency, and severity level. By automating the process of error detection and classification, the proposed system reduces manual effort and helps system administrators quickly identify recurring problems. This can support faster fault diagnosis, improve system monitoring, and contribute to better server reliability.

Signature of the Guide